



University of Fort Hare
Together in Excellence

PROCUREMENT DOCUMENT
FOR
FIRE PROTECTION INSTALLATION
FOR
FORT HARE UNIVERSITY FINANCE OFFICE
TENDER NUMBER: _____
DECEMBER 2025
(RETURNABLE DOCUMENT – SINGLE VOLUME)

Issued By:

Prepared By:

CA du Toit Eastern Cape (Pty) Ltd
22 Shirley Street,
Newton Park
Gqeberha,
6001

NAME OF TENDERER: _____

The Offered Total/s of the Prices inclusive of value added tax is:

Total from FINAL SUMMARY _____

Tenderer's CIDB CRS number _____

NOTE: The above is for recording of tender information purposes only. The Tenderer is required to fully complete C1.1: FORM OF OFFER AND ACCEPTANCE, which formally constitutes the tender offer submitted by the Tenderer, and failure to do so, could result in disqualification of the tender

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UNIVERSITY OF FORT HARE FINANCIAL OFFICE – FIRE PROTECTION

PART T1.5 – SCHEDULE OF EQUIPMENT OFFERED

Technical particulars and trademarks of all equipment offered in the tender (including equipment that has not specifically been specified, but will form part of the installation), must be supplied in this schedule.

The following information shall be provided in full and returned with the contractors' tender. Failure to comply with this condition of tender may invalidate the tender.

Tenderers shall attach to this page technical datasheets and product literature of the equipment offered.

PART C2: PRICING DATA

UNIVERSITY OF FORT HARE FINANCIAL OFFICE – FIRE PROTECTION

PART C2.1 – PRICING INSTRUCTIONS

2.1.1 PREAMBLE TO THE SCHEDULE OF PRICES

2.1.1.1 A lump sum price shall be provided.

2.1.1.2 The pricing schedule shall be used to evaluated progress claims.

2.1.1.3 The Contractor shall be required to provide a detailed cost breakdown of the individual lump sum prices to enable re-measurement of any changes and or/ variations.

2.1.1.4 All prices or rates inserted in the Pricing Schedule shall be EXCLUDING VAT. Provision has been made on the Summary Page, of the Pricing Schedule, for the addition of VAT.

2.1.1.5 Arithmetical errors of responsive tenders will be corrected in the following manner:

- Where there is a discrepancy between an amount shown in figures, and the corresponding amount stated in words, the amount stated in words shall take preference.
- In the Pricing Schedule, if there is an error in the line item total resulting from the product of the quantity and the unit rate, the line item total shall govern, and the rate shall be corrected.
- Where there is a misplacement of the decimal point in the unit rate, the line item total shall govern and the unit rate will be corrected.
- Where there is an error in the total of the prices either as a result of other corrections required by this checking process or in the tenderer's addition of prices, the total of the prices shall govern and the tenderer will be asked to revise selected item prices (and their rates in the Pricing Schedule) to achieve the tendered total of the prices.

Should a tenderer be unwilling to make the corrections ordered by the Engineer, the tender may be disqualified.

2.1.1.6 The units of measurement described in the Pricing Schedule are metric units. Abbreviations used in the Pricing Schedule are as follows :

mm	=	millimetre	hrs	=	hours
m	=	metre	kg	=	kilogram
km	=	kilometre	t	=	ton (1 000 kg)
m ²	=	square metre	No.	=	number
m ² .pass	=	square metre-pass	sum	=	lump sum
ha	=	hectare	MN	=	MegaNewton
m ³	=	cubic metre	MN.m	=	MegaNewton-metre
m ³ .km	=	cubic metre-kilometre	P C sum	=	Prime Cost sum
ℓ	=	litre	Prov sum	=	Provisional sum
kℓ	=	kilolitre	%	=	per cent
MPa	=	MegaPascal	kW	=	kilowatt

UNIVERSITY OF FORT HARE FINANCIAL OFFICE – FIRE PROTECTION

PART C2.2 – SCHEDULE OF PRICES

SCHEDULE 1: PRELIMINARY AND GENERAL ITEMS

Item	Description	Unit	Qty	Rate	Amount
1	Preliminary and General Items				
1.1	Compliance with the conditions of contract, contract data, preliminaries and general and all administrative costs and overheads.	Sum	1		R
1.2	Insurances	Sum	1		R
1.3	Construction Guarantee	Sum	1		R
1.5	Liaison with Main Contractor on programming and execution of works.	Sum	1		R
1.6	Liaison with other Contractors and Service providers (ie., Electrician, Medical Gas, Plumber, etc.) for final co-ordination between services with site management to ensure timeous and correct installation.	Sum	1		R
1.7	Contract Management and Site Supervision	Sum	1		R
1.8	Nett price to comply with all the requirements of the Health & Safety specification, ie Health and Safety file, Medicals, PPE for the duration of the contract.	Sum	1		R
1.9	Compilation of detail Operating and Maintenance Manuals, as-built drawings and documentation as specified and to the satisfaction of the Engineer. 3-off printed copies and 1x electronic version.	Sum	1		R
1.10	Training of the Employers staff.	Sum	1		R
1.11	12 months guarantee from date of Practical Completion Certificate.	Sum	1		R
1.13	12 months full maintenance and service of the installation in accordance with manufacturer's and suppliers specifications.	Sum	1		R
1.14	Any other P&G items: Contractor to specify	Sum	1		R
	TOTAL SCHEDULE 1 : CARRIED FORWARD TO FINAL SUMMARY				R

SCHEDULE 2: HVAC EQUIPMENT INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
2.1	Fire Hose Reel <u>Fire hose reels in accordance with SANS 543 and</u> <u>install in accordance with SANS 10105-2 and</u> <u>SANS 10400-W, complete with stop cock,</u> <u>Pressure gauge and gauge cock.</u>				
2.1.1	Supply	No.	4		
2.1.2	Install	No.	4		
3.1	Fire Extinguishers <u>Supply and install fire extinguishers in</u> <u>accordance with SANS 1910 complete with</u> <u>timber backing board, primed and painted with</u> <u>red gloss enamel finished coat.</u>				R -
3.1.1	4,5kg Dry Chemical Powder type	No.	2		
4.1	Pipework <u>Supply and install Galvanised steel piping to</u> <u>SANS 62-1 for supply to fire hose reels.</u> <u>Complete with degreasing, priming and painting,</u> <u>finished in 2-coats of red gloss enamel.</u>				
4.1.1	25mm Nominal Size	m	6	R	R
4.1.2	32mm Nominal Size	m	24	R	R
5.1	Pipe fittings <u>Supply and install Galvanised steel piping to</u> <u>SANS 62-2 for supply to fire hose reels</u>				
	90° Elbow				
5.1.1	25 x 25	No.	3	R	R
5.1.2	32 x 25	No.	1	R	R
5.1.3	32 x 32	No.	2	R	R
	Tee				
5.1.4	25 x 25 x 15	No.	1	R	R
	Union				
5.1.5	25mm	No.	1	R	R
5.1.6	32mm	No.	2	R	R
	Connection to mains				
5.1.7	<u>Complete with stainless steel 3-piece ball valve.</u> 32mm	No.	2	R	R
	Carried forward				R -

Item	Description	Unit	Qty	Rate	Amount
	Brought forward				R -
6.1	Signage				
	<u>Supply and install fire and directional signage in accordance with SANS 1186. Signs shall be wall mounted using corrosion resistant fixing</u>				
	Directional signs, 380x190, photoluminescent, single sided, framed				
6.1.1	EXIT	No.	4	R	R
6.1.2	Running Man/Down Arrow	No.	3	R	R
6.1.3	Running Man/Left Arrow	No.	1	R	R
	Fire signs, 380x190, photoluminescent, single sided, framed				
6.1.4	Fire Hose reel/Extinguisher	No.	4	R	R
	Fire signs, 570x190, photoluminescent, double sided, framed				
6.1.5	Fire Hose reel/Extinguisher/Left Arrow	No.	1	R	R
TOTAL FOR SCHEDULE NO. 2 :CARRIED FORWARD TO FINAL SUMMARY					R -

FINAL SUMMARY - MECHANICAL HVAC INSTALLATION

Schedule:

1	Subtotal Bill No: 1 - Preliminary and General	R	-
2	Subtotal Bill No: 2 - HVAC Equipment and Materials	R	-
	Sub-Total:	R	-
	Add : 5% Contingency	R	-
	Sub-Total	R	-
	15 % VAT	R	-
	TOTAL TENDER PRICE	R	-

NOTES TO BE READ IN CONJUNCTION WITH THE TENDER PRICE.

- 1 The amount given as the "Total Tender Price" and carried forward to Part C1.1 Form of Offer and Acceptance will be regarded as including the Provisional and Contingency amounts shown in the various schedules and no adjustments will be made for any failure by Tenderer's to include for any pricing items, sums or amounts in the total appearing on the official offer.

- 2 The entire mechanical installation in this contract shall be completed prior to completion and acceptance.

- 3 All Material offered should, wherever possible, be of South African manufacture with strong local (Port Elizabeth) after-sales service and backup. In the final adjudication of Tenders, preference will be given to bids offering such equipment and materials.

Name of Mechanical Contractor

Signature of Authorised Representative of Mechanical Contractor

Date

PART C3: SCOPE OF WORK

UNIVERSITY OF FORT HARE FINANCIAL OFFICE – FIRE PROTECTION

PART C3.1 – GENERAL TECHNICAL SPECIFICATION

The General technical requirements for the Fire Protection Installation will be in accordance with the Department of Public works PW371. This document is not included but can be obtained from the Department's website – www.publicworks.gov.za.

UNIVERSITY OF FORT HARE FINANCIAL OFFICE – FIRE PROTECTION

PART C3.2 – PROJECT TECHNICAL SPECIFICATION

1. SCOPE OF WORK

The scope of work is summarised below and includes, but not limited to, the supply, delivery, installation, testing and commissioning of the University of Fort Hare Financial Office – Fire Protection Installation

The scope of the fire protection services includes, but is not limited to the supply, delivery, offloading, installation and commissioning of Fire hose reels, Fire Extinguishers and Associated statutory signage as indicated on layout drawings and Bills of quantities

Associated Builders' Work

- All builders' work required shall be included and allowed for.

2. GENERAL

Where reference is made to Tenderer or Contractor in this specification the intent is to identify the specialist Fire Protection Contractor.

2.1. Statutory And Regulatory Requirements

The Installation shall be designed, erected, commissioned and maintained in compliance with the following applicable Acts and Regulations, as applicable:

The Occupational Health and Safety Act of 1993, as amended.

SANS 10400 2011: The application of the National Building Regulations.

Local Municipal by laws and regulations

Local Fire Regulations

Construction Regulations and Health and Safety Requirements

In addition the contractor shall exempt the Employer from all losses, costs or expenditures which may arise as a result of the contractor's negligence to comply with the requirements of the regulations enumerated in this Clause.

It shall be assumed that the contractor is conversant with the above-mentioned requirements. Should any requirement, by-law or regulation, which contradicts the requirements of this Document, apply or become applicable during erection of the Installation, such requirements, by-law or regulation shall overrule this Document and the contractor shall immediately inform the Engineer of such a contradiction.

All equipment and installations shall comply with the requirements of the Occupational Health and Safety Act 85 of 1993 and relevant SANS specifications. Where conditions are at variance, the Detail Technical Specification shall have preference over both General Technical Specifications and the Drawings.

Under no circumstances shall the contractor carry out any variations to the installation in terms of such contradictions without obtaining the written permission to do so from the Engineer.

The contractor shall supply and install all notices and warning signs that are required by the appropriate laws or regulations and/or by these Documents.

2.2. Reference Specifications and Standards

The latest revision of any Standard Specifications referred to in this Specification will be applicable.

Where a specification or standard is not specifically referred to it will be assumed that the relevant SANS, ISO, BSS, DIN or equivalent American standard, listed in order of preference, will apply.

The SI ("Le Système International d' Unités") Metric System of Units will apply. Refer to SANS M33A: The International Metric System: Guide to the use of the SI in South Africa.

All equipment and installations shall comply with the requirements of the Occupational Health and Safety Act 85 of 1993 and relevant SANS specifications.

Where conditions are at variance, the Detail Technical Specification shall have preference over both General Technical Specifications and the drawings.

2.3. Tender Drawings and Design Procedures

The tender design drawings are listed in Part C4.1 and attached in order to give an overview of the project.

These drawings provide sufficient information for pricing purposes, but do not necessarily indicate the exact position or detail the construction of equipment. Tenderers must satisfy themselves that the equipment offered by them will fit in the available space and can be positioned so that access for maintenance, cleaning, repair or removal is possible.

NOTE: All final dimensions must be determined on site before commencing the manufacture of any equipment. It is the responsibility of the contractor to verify all dimensions on site.

The contractor shall produce, and submit for approval, comprehensive, detailed, dimensional manufacturing (workshop) drawings of all sub-assemblies and components.

The contractor shall supply two (2) paper prints of all workshop drawings. The contractor shall allow one (1) week for drawing approval. The Contractor shall schedule design review session(s) with the Engineer and the Employer to facilitate input and/or clarification of the proposed detailed design. The Contractor shall update the drawings which shall then be submitted for signature. This will ensure that all prints used for construction will be certified as approved.

Any work done by the contractor without approved and signed drawings shall be completely at his own risk.

The contractor shall update all drawings on an on-going basis. Three (3) sets of final paper prints plus a CD copy of all the drawings in AutoCAD format, shall be supplied to the Engineer as part of the Operating and Maintenance Manual.

2.4. As-Built Drawings and Operation Instruction and Maintenance Manuals

Three (3) sets of final paper prints plus a CD copy of all the drawings in AutoCAD format, shall be supplied to the Engineer as part of the Operating and Maintenance Manual. Refer to the General Technical Specification for detail requirements.

It is stressed that the Completion Certificate will NOT be issued unless the O&M manuals are complete and approved by the Engineer.

2.5. Material and Workmanship

All materials shall be of the qualities specified and the contractor shall, upon request of the Engineer, furnish him with proof to his satisfaction that the materials are of the specified quality.

The contract works shall be executed in a workmanlike manner, to the satisfaction of the Engineer, and shall include all materials and equipment required for the successful operation of the installation as specified.

All materials and equipment used for the installation shall be new and undamaged.

The contractor shall, if requested by the Engineer, provide samples of material and equipment for approval. Such samples, if judged necessary by the Engineer, may only be returned after completion of the installation in order to ensure that the quality of the installed product is the same as that of the approved sample.

The contractor shall use only new materials of the best quality. A high standard of workmanship shall be maintained. The Engineer shall have the right to approve or disapprove any part of the installation if he deems it necessary in accordance with his professional judgement, interpretation of the Specification and accepted practice.

The contractor will be notified of any such disapproval of a part or parts of the installation and he will immediately rectify or replace that specific part or parts of the installation to the satisfaction of the Engineer. Any such instruction will not be deemed contrary to the Contract and will not relieve the contractor of his responsibility to complete the installation on the originally agreed completion date.

2.6. Maintenance and Servicing During The 12 Month Guarantee Period

The contractual guarantee period on faulty equipment, materials and workmanship shall be twelve (12) months. The guarantee period on material, equipment and labour commences on the date of the completion certificate issued by the Engineer and expires twelve (12) months later.

During the guarantee period, the contractor shall be responsible for the complete maintenance and servicing of equipment and plant according to the suppliers/ manufacturer's specifications. Maintenance of the installation shall mean the regular servicing, lubrication, repairing, cleaning and adjustment of the installation as recommended by the manufactures as well as the free of charge replacement of any defective components during the guarantee period.

A suitably qualified and trained person shall routinely and regularly examine and test the installation once every month and shall also perform all the necessary maintenance tasks to ensure smooth and faultless operation. A quarterly report shall be submitted to the Engineer.

The contractor shall immediately, on the day of first call-out, attend to breakdown/emergency calls. In the event of non-performance by the contractor in this respect, the owner shall be entitled to make such other arrangements as are necessary, the cost of which shall be for the contractor's account or deductible from any outstanding retention monies.

In the event of a call-out by the Engineer during the one (1) year maintenance period the contractor shall make available a qualified technician to attend to any plant related problems within 4-hours (from time of instruction by Engineer to arrival on site). The Engineer may require emergency response in which case the Contractor shall take such action that is necessary to ensure an immediate response. The fault(s) shall be corrected and a report issued to the Engineer on the exact detail of the breakdown and the corrective steps taken to prevent re-occurrence of a similar fault.

A logbook shall be kept and all servicing and repairs shall be recorded in this logbook with meticulous care. The logbook shall at all times be put at the disposal of the Engineer. The contractor shall issue the logbook with full record of all services and repairs to the owner after the guarantee period has expired.

2.7. Training

The Tenderer shall include for detail instruction and training sessions with the Employer's technical department.

2.8. Make of Equipment

The contractor will be allowed to offer equivalent equipment for consideration, under the following explicit conditions:

The equipment shall comply in all respects with the requirements specified and shall be subject to approval by the Engineer.

2.9. Equipment shall be of equivalent quality and performance.

The delivery period for such equipment shall not jeopardise the construction programme.

The local manufacturer or supplier shall have adequate service facilities and/or keep sufficient stocks of spares required for servicing.

The equipment shall be offered as an ALTERNATIVE if offered at tender stage with the accompanying cost saving.

The contractor shall supply such information regarding the equipment as required by the Engineer.

2.10. Inspections And Testing

Inspections (Part III, SAACE 1978)

The Engineer shall have general supervision and give direction to the contract works. Supervision shall comprise such periodic visits, as the Engineer may consider necessary to inspect the contract works for conformity with the contract documentation and to provide clarification and further information as necessary.

The Engineer shall have the power at any time to inspect and examine any part of the contract works or any materials intended for use in or on the contract works, either on the site or at any factory, workshop or other place where such parts or materials are being constructed manufactured or at any place where same are lying or from where they are being obtained, and the Contractor shall give all such facilities as the Engineer may reasonably require to be given for such inspection and examination.

The Contractor shall not be liable for the cost of inspecting materials at the place of manufacture, construction or storage nor any travelling or accommodation costs arising out of the execution of such inspection etc.

2.11. Alternative Equipment

The words "similar or equal approved" is always implied whether following the description of equipment, or not.

Should names of equipment manufacturers appear in the schedules, those are around which the design has been based and all equipment so scheduled is complementary.

The tenderer may offer alternative makes providing that such alternatives are of equal quality to those "selected" by the Engineer and should the tender be accepted the approval of the Engineer is obtained before ordering.

The Engineer shall be the sole judge of the equipment offered and his decision shall be final.

As stated above all equipment selections are complementary and it is the responsibility of the successful tenderer to ensure that: -

His alternative offer can be housed within the space provided, leaving sufficient clearance surrounding the unit as laid down by local regulations and for ease of maintenance. Should the alternative have to be replaced by that as listed in order to meet these requirements, it shall be at no additional cost to the Contract.

All openings in the structure meet with the requirements of his alternative offers, or ensure that timely details of any relevant amendments are requested to prevent any abortive work on site.

If there are any monetary benefits which can be offered to the Client through an alternative item of equipment, the Tenderer must incorporate all secondary cost variations into that saving reflected (i.e. the price reduction offered for an alternative chiller shall incorporate any additional costs in upgrading a pump and/or its motor size which could result from an increase in resistance through its heat exchangers).

Further, all relevant technical data, especially regarding energy consumption, rated output, capacity, variations in resistances, etc., must also be submitted in order that the Engineer may evaluate savings in capital costs against any increases in operating costs prior to accepting that particular alternative offer.

2.12. Hoisting, Rigging and Scaffolding

All hoisting, rigging and scaffolding required/associated with the Works described by this specification shall be allowed for by the Fire Protection Subcontractor.

All scaffolding and elevated platforms to meet with the requirements of the OHS Act, Construction Regulations and Health and Safety procedures.

2.13. Quality of Materials

Only materials of first-class quality shall be used and all materials shall be subject to the approval of the Representative/Agent or the Client.

Wherever applicable the material is to comply with the relevant South African Bureau of Standards (SANS/SABS), specifications, or to NFPA /UL/FM Standards, where no SABS/SANS Specifications exist.

Materials wherever possible, must be of South African manufacture.

Wherever a trade name for any product has been described in the bills of quantities / lump sum document, the tenderer's attention is drawn to the fact that any other product of equal quality may be used subject to the written approval of the principal agent being obtained prior to the closing date for submission of tenders

If prior written approval for an alternative product is not obtained, the product described shall be deemed to have been tendered for.

2.14. Testing

The Contractor shall supply all test equipment, test facilities and everything necessary at his cost, to perform these tests.

The Engineer shall be notified one (1) week in advance of any tests so that he may witness such tests.

Pressure testing shall be carried out in accordance with the following parameters. Tests shall be recorded and witnessed and test reports shall be made available on requested by the engineer

Test medium	Test pressure	Duration of test
Air	800 kPa	24 hours
Water	800 kPa	24 hours
Water	2000 kPa	1 hour

Items incapable of withstanding 20 Bar pressure test shall suitably isolated during testing.

After completion of the works and before first delivery is taken, a full test will be carried out on the installation for a period of sufficient duration to determine the satisfactory working thereof. During this period the installations will be inspected, and the Contractor shall make good, to the satisfaction of the Representative/Agent, any defects which may arise.

The Contractor shall provide all instruments and equipment required for testing and any water, power and fuel required for the commissioning and testing of the installations at completion.

2.15. Piping:

Pressure test of fire water piping at a test pressure of 1,5 times the maximum working pressure at the lowest point in the system, but not less than 1800kPa.

All instrumentation or other equipment, which could be damaged during the pressure test, shall be removed from the pipe systems. The relevant system shall be filled with water and all high points shall be vented at least 24 hours before the test.

The duration of the pressure test shall be 2 hours, after which no water leaks shall be visible and no pressure drop shall occur after corrections have been made for changes in ambient temperature during the test period.

Pipework

All pipework shall be hot dipped galvanised steel.

Item	Joining Method
Pipework of nominal diameter up to and including 50mm	Threaded

All sprinkler pipework concealed or exposed either above ground or in-ground shall be suitably cleaned, primed and painted with a finished coat in accordance with SANS 10140- 3.

All pipework shall, prior to delivery to site, be primed and painted with finished coat.

Pipes shall be protected, for example, by caps against the entry of foreign matter, such as rubble and cordage, and shall be examined for such matter before erection.

All pipework where buried, encased in concrete or penetrating concrete or brickwork shall be suitable wrapped with denso tape in the affected areas.

No pipework shall exceed 6m in length without either threaded, mechanical or flanged joint.

General application of joints

Item	Joining Method
Pipework of nominal diameter up to and including 50mm	Threaded

Threaded Couplings & Joints

Threaded fittings shall be malleable cast iron with hot dipped galvanised finish.

All threaded fittings shall be in accordance with SANS14.

Pipe Supports

Pipe supports shall be of type accepted by ASIB for use on sprinkler systems.

The following maximum distance between pipe supports shall be adhered to

Pipe nominal bore (mm)	Maximum support spacing (mm)
25	1800
32	2000
40	2100
50	2400
65	2700
80 - 150	3000

If leaks are found, welded connections shall be cut out and re-welded and screwed joints shall be dismantled, cleaned and re-connected. Rectified piping shall be re-tested.

Painting

All piping, including pipe fittings, couplings and flanges shall be primed and painted prior to installation.

Metal surfaces shall be prepared using a suitable degreaser, followed by one coat of etching primer and two coats of enamel paint (Signal Red).

The following paints, or similar approved shall be used for coating of pipework.

Degreaser	
- Galvanised - Plain	Plascon Metalcare Galvanised Iron Cleaner

	Plascon Metalcare Waterbased Degreaser
Primer	Plascon Metalcare Etch Primer
Final coat	Plascon Super Universal Enamel (Signal Red: G7)

The manufacturer's guideline shall be adhered to in the surface preparation and application of coatings.

2.16. Pressure Gauges

All pressure gauges fitted to a sprinkler system shall:

Read in kPa.

Have a maximum scale value of 150% of the maximum working pressure of the system.

Have divisions not exceeding 20 kPa for pressures up to 1000 kPa.

Have divisions not exceeding 50 kPa for pressures above 1000 kPa and up to 1600 kPa.

Have divisions not exceeding 100 kPa for pressures above 1600 kPa.

Have a dial face of not less than 100 mm in diameter.

Be of the glycerine filled type.

Be installed complete with 3-way gauge cock with bleed to be able to confirm gauge operation back to zero and enable each pressure gauge to be readily removed without interruption of the installation water supplies.

2.17. Hose Reels

Any hose reel installed in such building shall comply with the requirements in SANS 543, shall be installed in accordance with SANS 10105-1 and SANS 10400-W.

All hose reels installed shall bear, in a prominent position on the reel disc facing the user, a certification mark from an accredited certification body.

All hose reel drums, including inlet pipe and mounting brackets, shall be of stainless steel construction and the hose of PVC in accordance with SANS 1086.

2.18. Fire Extinguishers

All portable fire extinguishers installed with the requirements in SANS 1910, and shall be installed, maintained and serviced by competent persons in accordance with SANS 1475-1 and SANS 10105-1.

Such portable fire extinguishers shall bear a certification mark from an accredited certification body.

2.19. Evacuation Signage

All evacuation and safety signage shall be of the photoluminescent type and comply with the requirements of SANS 1186-1 & SANS 1186-5.

Signage shall be mounted on a 1,2mm thick aluminium backing which shall be wall mounted using suitable anchor bolts.

2.20. Testing

The contractor shall perform interim pressure tests during the sprinkler system during installation.

Fire services shall be tested using:

Test medium	Test pressure	Duration of test
Air	800 kPa	24 hours
Water	800 kPa	24 hours
Water	2000 kPa	1 hour

Items incapable of withstanding 20 Bar pressure test shall suitably isolated during testing.

After completion of the works and before first delivery is taken, a full test will be carried out on the installation for a period of sufficient duration to determine the satisfactory working thereof. During this period the installations will be inspected, and the Contractor shall make good, to the satisfaction of the Representative/Agent, any defects which may arise.

The Contractor shall provide all instruments and equipment required for testing and any water, power and fuel required for the commissioning and testing of the installations at completion.

PART C4: ANNEXURES

UNIVERSITY OF FORT HARE FINANCIAL OFFICE – FIRE PROTECTION

APPENDIX A: DRAWINGS

Drawing Register:

DRAWING NO.	TYPE	DESCRIPTION
P8080M07-001	A1	FIRE AND WAYFINDING LAYOUT